

EVALUATION AND ADAPTATION TO FREQUENCY-TO-PLACE FACTORS IN COCHLEAR IMPLANT USERS

World Congress of Audiology
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INTRODUCTION

Individual factors

age at onset of hearing loss, aetiology,
duration of moderate and severe-to-
profound hearing loss...



Variability in performances

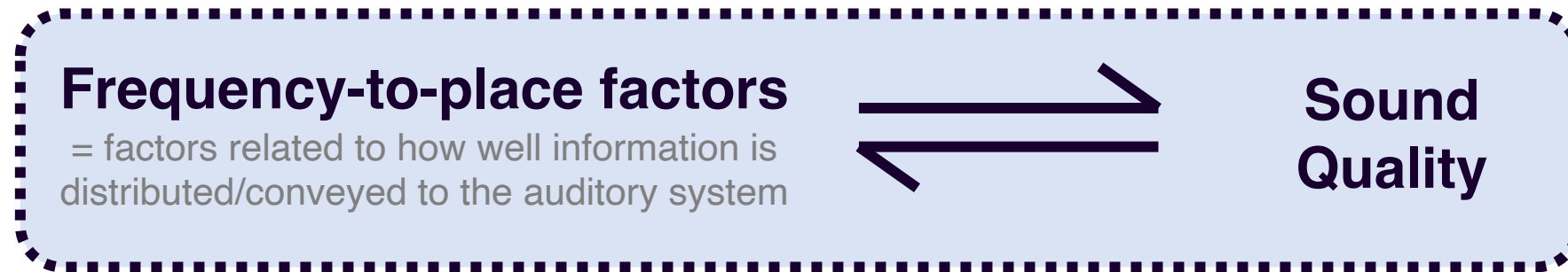
Speech in noise, sound quality, quality of
life..



INTRODUCTION

My goal

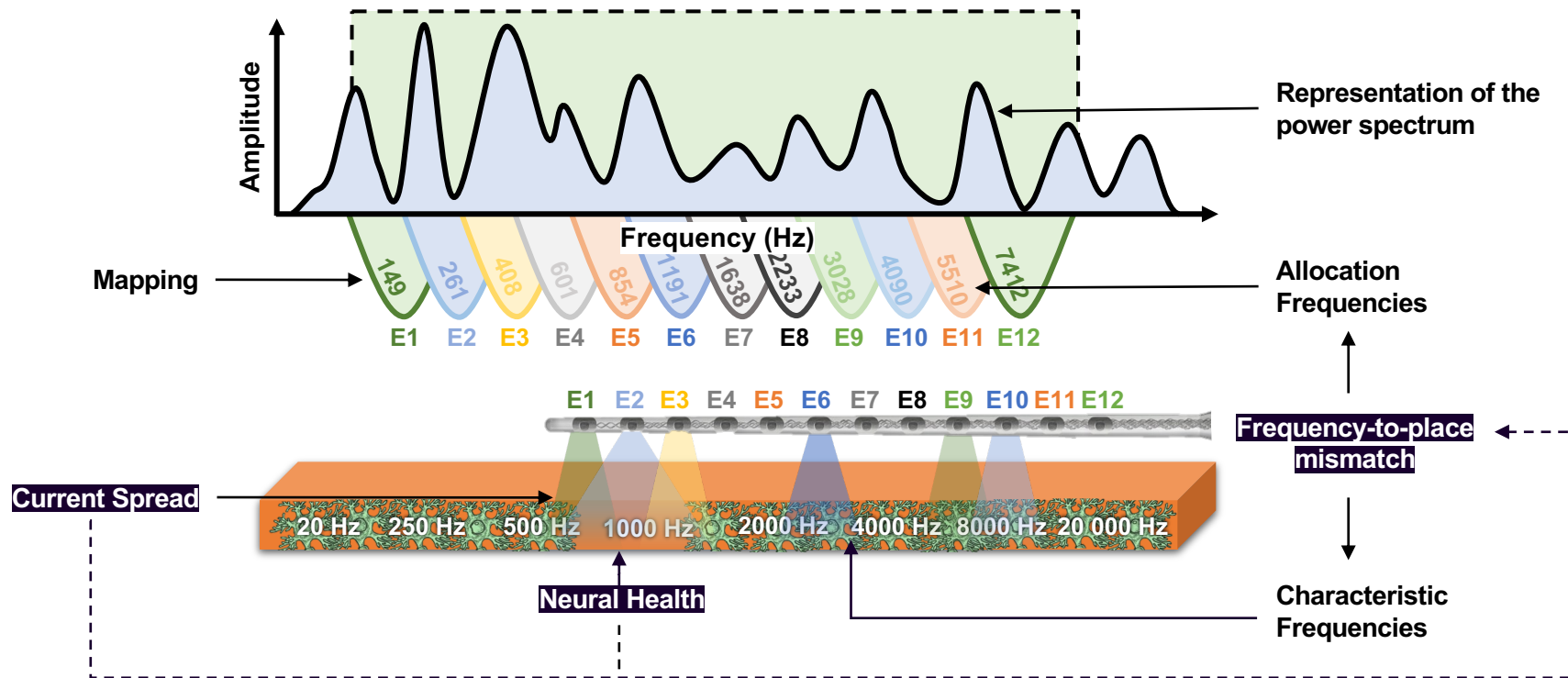
Select and study factors with the potentiality to be manipulated



INTRODUCTION

Frequency-to-place factors

= factors related to how well information is distributed/conveyed to the auditory system



INTRODUCTION

FTPM & electrode specificity



Effect of frequency-to-place mismatch on speech and music sound quality in acoustic cochlear implant simulation

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ABSTRACT:

Sound quality perception for cochlear implant (CI) users has become increasingly important. Although many CI users achieve near-normal speech recognition in quiet, they often report poor sound quality, particularly for music. One factor contributing to this degradation is frequency-to-place mismatch (FTPM), which occurs when electrode positions do not align with the cochlea's characteristic frequency map. This study aimed to better understand the impact of FTPM on sound quality in CI simulations across different signal types and configurations. Twenty-three normal-hearing participants were tested (online or onsite) using an adapted Multi Stimulus test with Hidden Reference and Anchor (MUSHRA) paradigm. Different FTPM configurations were simulated with a noise vocoder to assess their influence on speech and music sound quality. **Results showed that greater FTPM caused noticeable degradation, especially when lower frequencies were affected. Variability in FTPM across electrodes also significantly reduced perceived quality. Furthermore, the impact of FTPM depended on the type of stimulus, with speech and music showing distinct sensitivity patterns.** Online assessments closely matched onsite results, confirming the reliability of remote testing. Together, these findings clarify why sound quality perception differs between CI users and contexts and highlight new opportunities to develop strategies for alleviating mismatch effects in CIs. © 2026 Acoustical Society of America. <https://doi.org/10.1121/10.0043470>

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Electrodes affected by FTPM matter!

*Mismatch Variability between electrodes
Low Freq mismatch > High Freq mismatch*



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INTRODUCTION

Goals

Goal 1

Develop tests at the electrode level

Able to track real perception challenges at the electrode level

Goal 2

Link them to our factors

FTPM, Current Spread and Neural Health, for different CI user populations and see if we are able to evaluate the effect of these factors on perception



INTRODUCTION

Testing Methods

Estimation of internal representation of vowels in experienced cochlear implant users and its relationship with frequency-to-place factors

Louis Villejoubert¹, Lorenzo Picinali², Kathleen Faulkner³, Marina Salorio-Corbetto¹, Susan Eitutis⁴, and Deborah Vickers¹

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Experienced users

Speech related task

Frequency importance function of cochlear implant users derived from music-related task: relationship with frequency-to-place mismatch, neural responsiveness and current spread

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Experienced users

Music related task

Adaptation to frequency-to-place factors in cochlear implant users during the first months of cochlear implant use: longitudinal case study of vowel and chord perception

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Newly implanted

Speech + Music related task



METHODS

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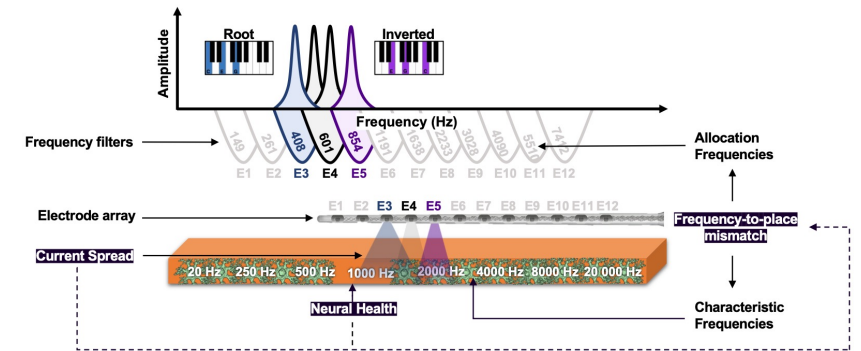
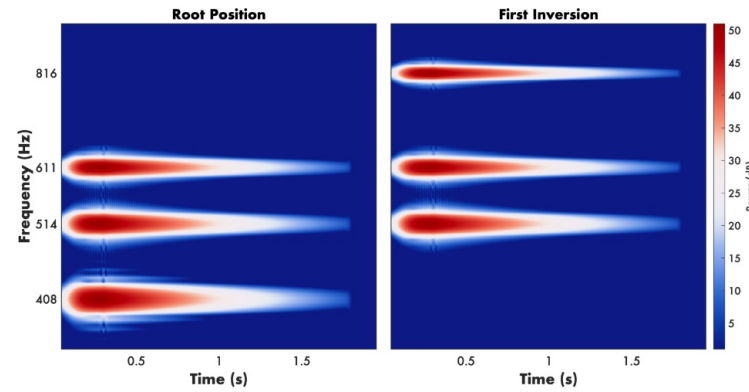
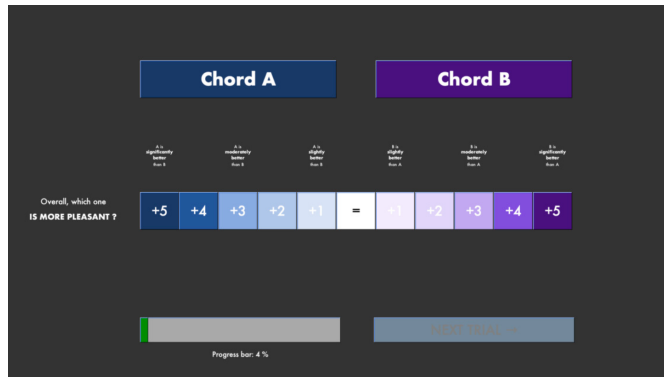


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METHODS

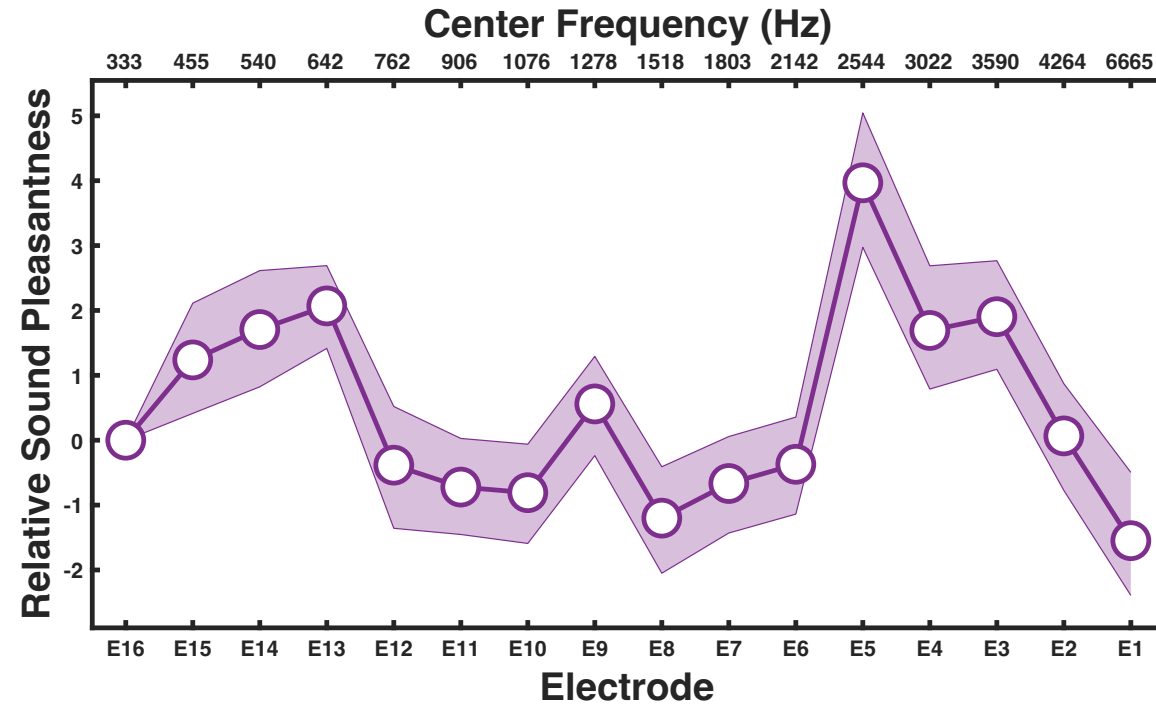
Idea of the task

Concept of inverted chords



METHODS

Frequency importance function



METHODS

**14 CI users
implanted as adults,
after language
acquisition**

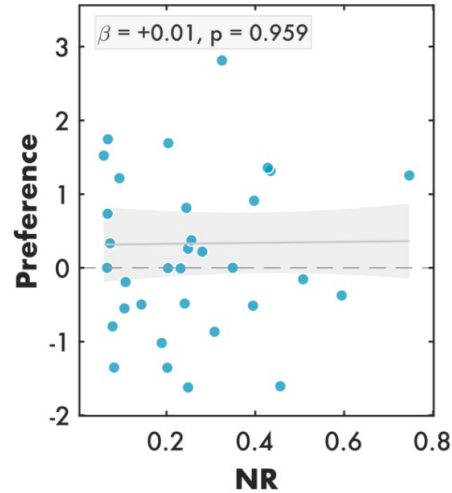
**10 CI users
implanted as
children, before
language
acquisition**



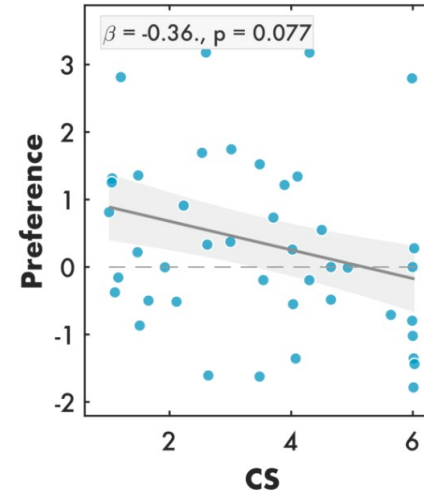
RESULTS

14 CI users
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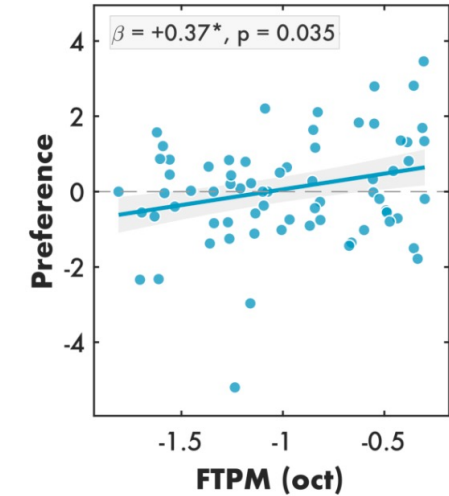
Neural Health



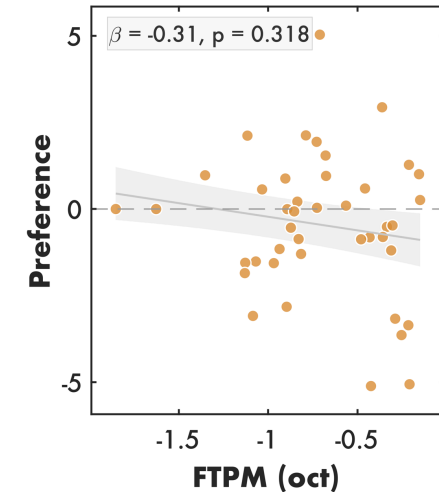
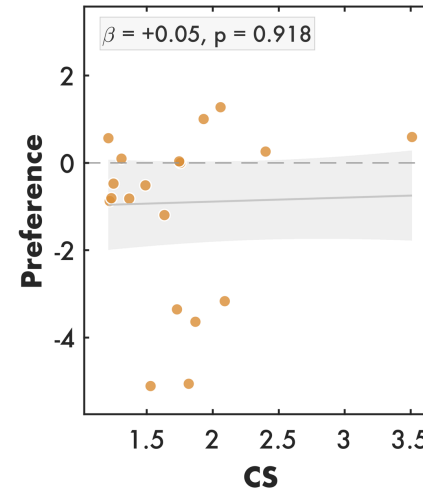
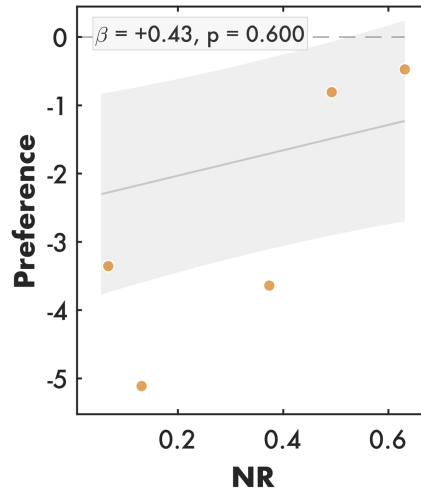
Current Spread



Frequency-to-place
mismatch



10 CI users
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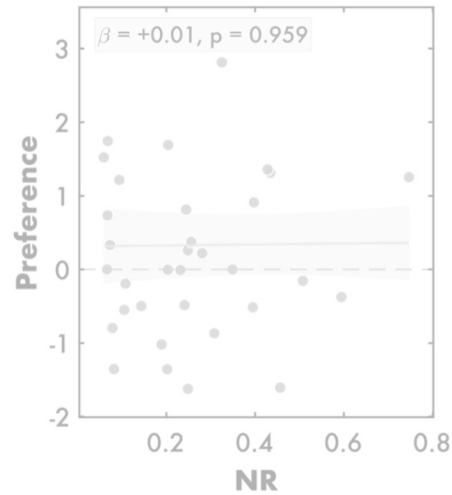


RESULTS

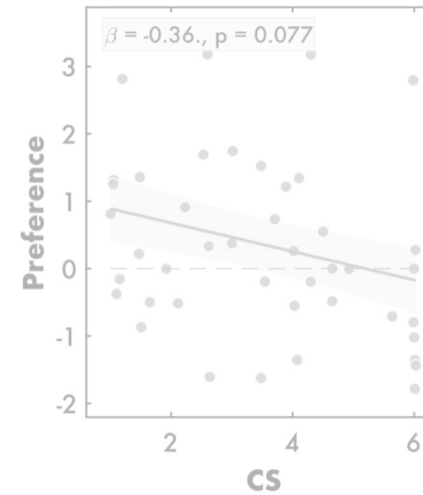
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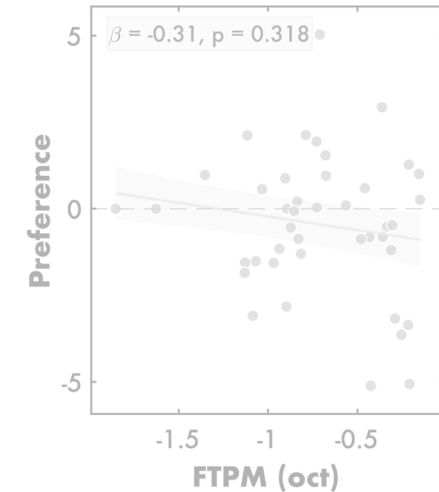
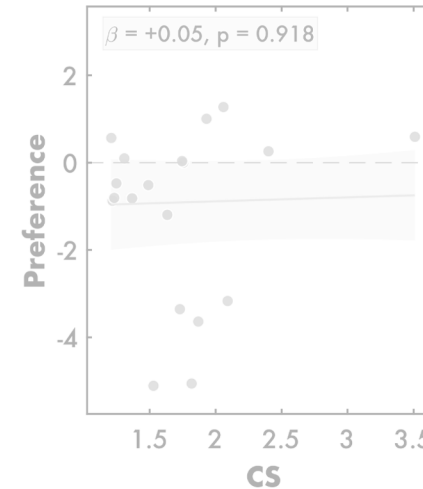
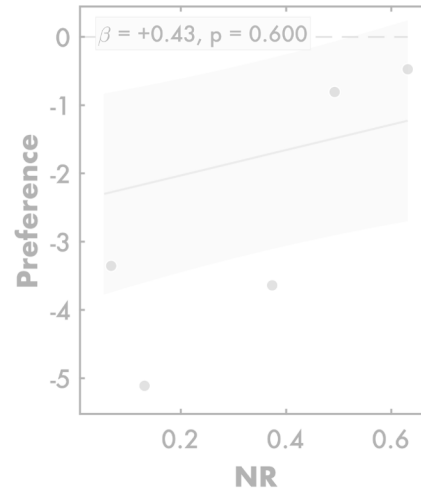
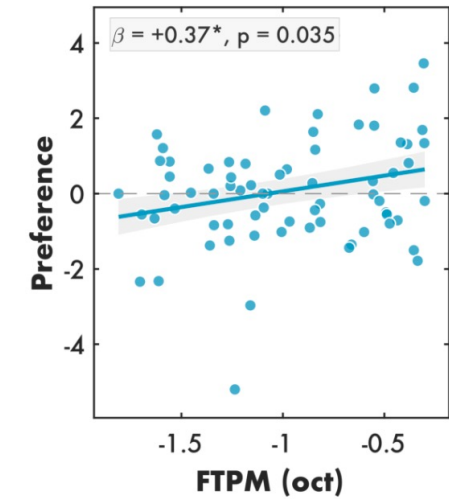
Neural Health



Current Spread



Frequency-to-place mismatch



CONCLUSION?

Long term effect

Frequency-to-place
mismatch

Yes

Need re-mapping?

Current Spread

Resolved?

Neural Health

Resolved?

How we ended up here?



RESULTS

Adaptation process

The switch-on participant

Early

Extended

Day +6 → Day +20 → Day +43 → Day +134

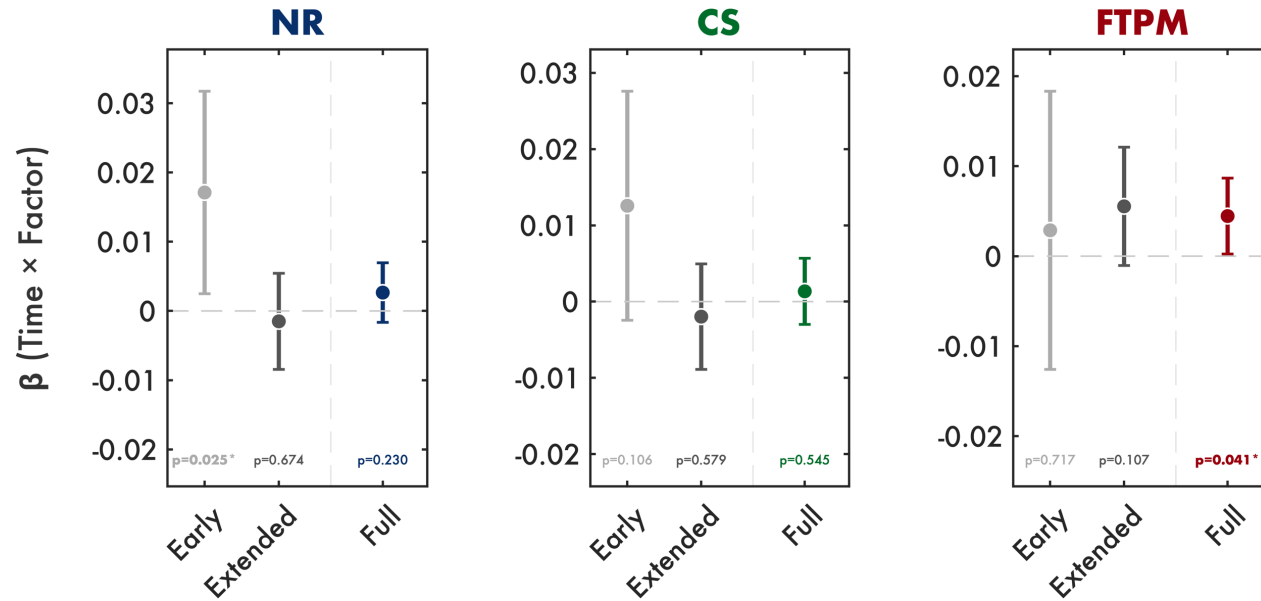
post switch-on



RESULTS

Adaptation process

The switch-on participant



Early adaptation to NR

**Late adaptation to
FTPM**



TAKE AWAY AND OPEN QUESTIONS

Evaluate at the individual electrode sites is crucial

if we want to propose targeted re-mapping strategies to clinicians

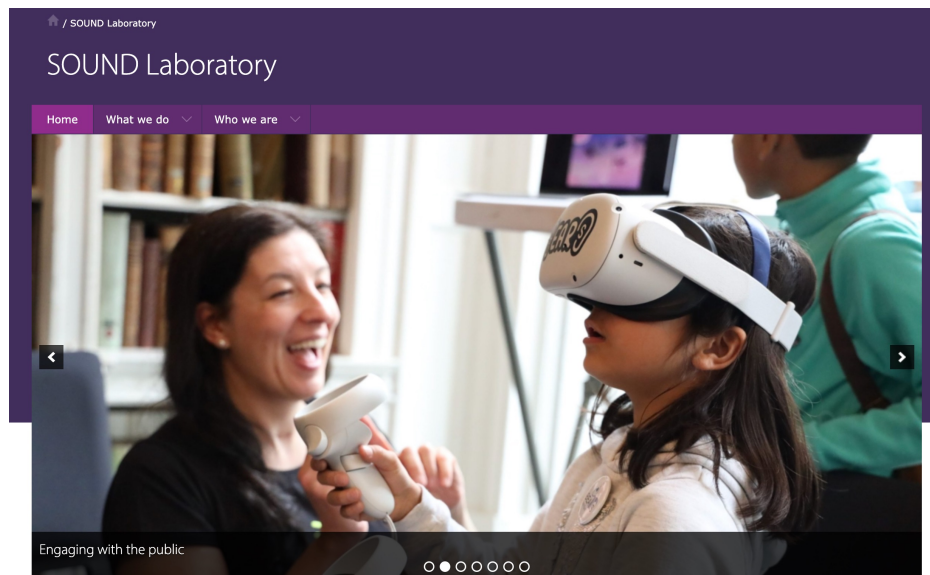
Neural Responsiveness component is resolved early in the adaptation process

Electrode associated with sharp responses (*ie*, good NR) are probably the key. We found that they are associated with lower preferences at the early stages of the adaptation.. What would be the implications for care?

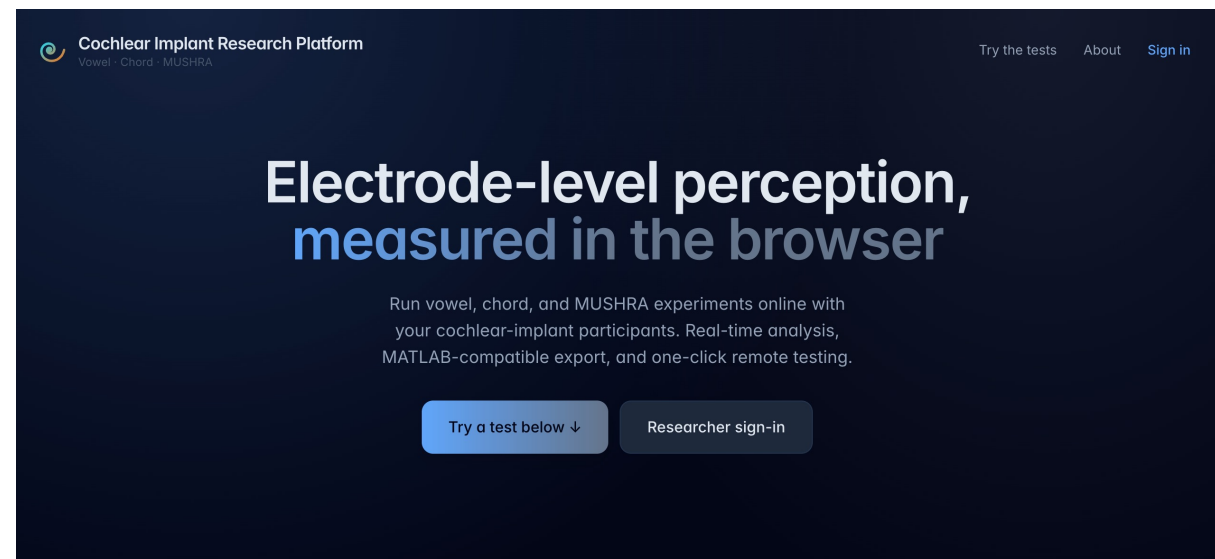
FTPM still matters after years of CI use for our music task

Targeted re-mapping even for experienced users might improve sound quality.. How? "Flat FTPM"?





<https://sound-lab.medschl.cam.ac.uk/>



<https://very soon website.cam.ac.uk/>

Thank you for your attention



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